

Sakar Robotics

Keenon C40 S - Postman cURL Testing & Control Reference

Prepared from the supplied Keenon Open Platform V2.4 documentation and the live API responses provided during testing

Important security note: Access tokens, client secrets, and credentials are intentionally represented as placeholders. Do not place real Keenon credentials in frontend code or distribute them to clients.

1. Test Environment and Robot

Item	Value
Keenon Store ID	C00715655
Store Name	Sakar robotics office
Robot Name	Demo Piece
Robot ID / Robot SN	94:BA:06:CA:99:F3
Keenon Model	C40 S (internal integration value)
Sakar Product Name	Sakar CleanBot 5000 Plus
Scene	dTW2N7 - SR Cleaning
Map ID	4c0075859805496eb452187b3cd91107
Charging Point	39 - 1_Charging pile

2. Live Cleaning Area Mapping

Area	Current Area ID
Conference carpet	141af8448ffc4ddaa204259c02456eb0
Work area	5d5bc7a20f984f45ada7450fac4be972
Lobby	5efd6bfea62b4fc29c425bd36997071a

Important: Area IDs are live configuration values. Do not hardcode them permanently in the Sakar platform; sync the current area list and maintain a mapping to Sakar site/area records.

3. Cleaning Modes Confirmed

Mode ID	Mode
101	Sweep & Mop
102	Water Suction
103	Sweep & Vacuum
104	Sweep & Push
105	Sweep

4. Authentication - Postman cURL

```
curl --location '{{base_url}}/api/open/oauth/token' \
--request POST \
--header 'Content-Type: application/x-www-form-urlencoded' \
--header 'Accept: application/json' \
--data-urlencode 'client_id={{client_id}}' \
--data-urlencode 'client_secret={{client_secret}}' \
--data-urlencode 'grant_type=client_credentials'
```

In Postman, store the returned access_token as the environment variable **token**.

5. Store and Robot Data

Get Stores

```
curl --location '{{base_url}}/api/open/data/v1/store/list' \
--header 'Authorization: Bearer {{token}}' \
--header 'Accept: application/json'
```

Get Robot List

```
curl --location '{{base_url}}/api/open/data/v1/store/robot/list?storeId={{store_id}}' \
--header 'Authorization: Bearer {{token}}' \
--header 'Accept: application/json'
```

Robot Status

```
curl --location '{{base_url}}/api/open/scene/v1/robot/status?robotId={{robot_id}}' \
--header 'Authorization: Bearer {{token}}' \
--header 'Accept: application/json'
```

Battery

```
curl --location '{{base_url}}/api/open/custom/robot/battery/level?robotSn={{robot_sn}}' \
--header 'Authorization: Bearer {{token}}' \
--header 'Accept: application/json'
```

Cleaning Status

```
curl --location '{{base_url}}/api/open/custom/clean/robot/status?robotSn={{robot_sn}}' \
--header 'Authorization: Bearer {{token}}' \
--header 'Accept: application/json'
```

6. Cleaning Configuration APIs

Area List

```
curl --location '{{base_url}}/api/open/custom/clean/robot/area/list?storeId={{store_id}}&robotSn={{robot_sn}}&currentPage=1&pageSize=5' \
--header 'Authorization: Bearer {{token}}' \
--header 'Accept: application/json'
```

Cleaning Modes

```
curl --location '{{base_url}}/api/open/custom/clean/robot/strategy/clean/model?robotSn={{robot_sn}}' \
--header 'Authorization: Bearer {{token}}' \
--header 'Accept: application/json'
```

Return / Charging Points

```
curl --location '{{base_url}}/api/open/custom/clean/robot/strategy/back/point?robotSn={{robot_sn}}' \
--header 'Authorization: Bearer {{token}}' \
--header 'Accept: application/json'
```

Cleaning Logs

```
curl --location '{{base_url}}/api/open/custom/clean/log/list?storeId={{store_id}}&robotSn={{robot_sn}}&currentPage=1&pageSize=5' \
--header 'Authorization: Bearer {{token}}' \
--header 'Accept: application/json'
```

7. Cleaning Task - Tested Successfully

The following temporary task parameters were proven against the supplied live robot. Conference carpet uses the current area ID shown above; Sweep is mode 105; charging point is 39.

```
curl --location '{{base_url}}/api/open/custom/clean/robot/strategy/temporary/task' \
--request POST \
--header 'Authorization: Bearer {{token}}' \
```

```
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data '{
  "robotSn": "{{robot_sn}}",
  "areaIdList": [
    "{{conference_area_id}}"
  ],
  "cleanModelId": "{{clean_model_id}}",
  "cleanTimes": 1,
  "backPointId": "{{back_point_id}}"
}'
```

Expected API receipt: code 610000 with bizType **CleanStrategyTemporary**. A 610000 receipt means the request was accepted; the task should be verified through status/logs and not marked complete solely from the receipt.

8. Lobby Sweep - Current Live Area ID

```
curl --location '{{base_url}}/api/open/custom/clean/robot/strategy/temporary/task' \
--request POST \
--header 'Authorization: Bearer {{token}}' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data '{
  "robotSn": "{{robot_sn}}",
  "areaIdList": [
    "{{lobby_area_id}}"
  ],
  "cleanModelId": "{{clean_model_id}}",
  "cleanTimes": 1,
  "backPointId": "{{back_point_id}}"
}'
```

9. Stop / Finish Current Cleaning Task

```
curl --location '{{base_url}}/api/open/custom/clean/robot/finish/task' \
--request POST \
--header 'Authorization: Bearer {{token}}' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data '{
  "robotSn": "{{robot_sn}}"
}'
```

10. Pause Current Cleaning Task

```
curl --location '{{base_url}}/api/open/custom/clean/robot/pause/task' \
--request POST \
--header 'Authorization: Bearer {{token}}' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data '{
  "robotSn": "{{robot_sn}}"
}'
```

11. Return to Charging Station

```
curl --location '{{base_url}}/api/open/custom/clean/robot/recharge/task' \
--request POST \
--header 'Authorization: Bearer {{token}}' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data '{
  "robotSn": "{{robot_sn}}"
}'
```

This recharge endpoint was live-tested successfully for the same robot and returned code 610000 / CleanRobotRechargeTask.

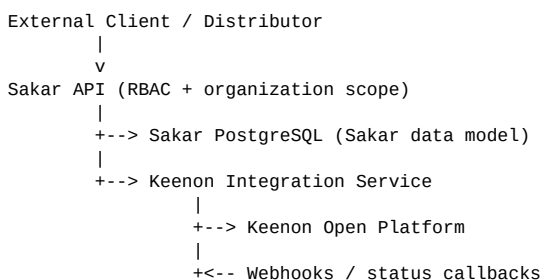
12. Verification and Troubleshooting Flow

1. GET /store/robot/list
2. GET /scene/v1/robot/status
3. GET /custom/clean/robot/status
4. GET /custom/clean/robot/area/list
5. GET /custom/clean/robot/strategy/clean/model
6. GET /custom/clean/robot/strategy/back/point
7. POST temporary cleaning task
8. Watch /custom/clean/robot/status
9. Check /custom/clean/log/list
10. Verify final mState / failDesc / cleanArea / cleanTiming

13. Important Live Test Findings

Finding	Result
Store and robot list	Working; store C00715655 contains the tested C40 S robot.
Sweep mode	Mode 105 confirmed.
Recharge command	Confirmed accepted with 610000 / CleanRobotRechargeTask.
Temporary cleaning command	Confirmed accepted with 610000 / CleanStrategyTemporary.
Successful Lobby cleaning	Confirmed in logs: cleanArea 13.24, cleanTiming 229 sec, mState 1, failDescCode 0.
Earlier Conference carpet issue	Initial area ID was stale/wrong; current area mapping provides a new Conference carpet ID.
Historical supply failure	Repeated logs showed failDesc 467 / "■■■■■■" (no clean water added).
Current state semantics	V2.4 defines mainState 3 = Work and subState 42 = Cleaning; subState 44 = Returning.

14. Sakar Architecture Recommendation



External users should see:

- Sakar CleanBot 5000 Plus
- Sakar robot ID
- Sakar site / area names

Keenon-specific identifiers and credentials stay inside Sakar.

For the production platform, synchronize Keenon robot/area configuration into Sakar, persist task history and webhook events in Sakar, and expose Sakar-owned APIs to clients. The supplied audit recommends a cleaning-first backend and webhook-based live state with conservative polling as reconciliation.

15. Current Postman Variable Set

```
base_url      = https://cloud.robotkeenon.com
token         = <set after OAuth>
client_id     = <secret>
client_secret = <secret>
store_id      = C00715655
robot_id      = 94:BA:06:CA:99:F3
robot_sn      = 94:BA:06:CA:99:F3
scene_code    = dTW2N7
map_id        = 4c0075859805496eb452187b3cd91107
back_point_id = 39
clean_model_id = 105
conference_area_id = 141af8448ffc4ddaa204259c02456eb0
work_area_id  = 5d5bc7a20f984f45ada7450fac4be972
lobby_area_id = 5efd6bfea62b4fc29c425bd36997071a
```